

Starting with the terraform init:

```
joaoibeiro@JOACAPC:~/terraform-lab-azure$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 3.0"...
- Installing hashicorp/azurerm v3.117.1...
- Installed hashicorp/azurerm v3.117.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

Then Terraform plan:

```
joaoibeiro@JOACAPC:~/terraform-lab-azure$ terraform plan
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# azure_rm_linux_virtual_machine.vm_demo will be created
+ resource "azure_rm_linux_virtual_machine" "vm_demo" {
  + admin_username           = "joaolab"
  + allow_extension_operations = true
  + bypass_platform_safety_checks_on_user_schedule_enabled = false
  + computer_name            = (known after apply)
  + disable_password_authentication = true
  + disk_controller_type     = (known after apply)
  + extensions_time_budget   = "PT1H30M"
  + id                      = (known after apply)
  + location                 = "spalincentral"
  + max_bid_price            = -1
  + name                    = "vm-tf-lab-joao"
  + network_interface_ids    = (known after apply)
  + patch_assessment_mode    = "ImageDefault"
  + patch_mode               = "ImageDefault"
  + platform_fault_domain    = -1
  + priority                 = "Regular"
  + private_ip_address       = (known after apply)
  + private_ip_addresses     = (known after apply)
  + provision_vm_agent       = true
  + public_ip_address        = (known after apply)
  + public_ip_addresses     = (known after apply)
  + resource_group_name      = "rg-tf-lab-joao"
  + size                     = "Standard F2"
  + virtual_machine_id       = (known after apply)
  + vm_agent_platform_updates_enabled = false

  + admin_ssh_key {
    + public_key = <<EOF
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCA4e0Z94sDNGur1G1SxOK1YX0xhEFT1e5rR1e1PnsuclHm5T5KsWcQp8m3SdqU3tEuXgLB3GQQAeAB8XQD0JNGCpPYcYxpGuY08XotIDFxmJRo
3tFahm2fKf0E8L8SL8eap9SP1cy1nQ4DMpzyDABHLUG6tyqueUOXUFTQ71Q4GfU8ANyM6/N1211EAy1SncHMu1FPqArCaFFZce4wC112Em08eJy/ThrJ7Fov8B1mu/or99VLN72gS21KPos1BrSSJ045frikqHubu
iFTgr5uQDqG6GJ0J190MyE1vXkL7IV9tCOT8wMky5SdbJppkGMP/ joaoibeiro@JOACAPC
EOF
    + username   = "joaolab"
  }
}

+ os_disk {
  + caching           = "ReadWrite"
  + disk_size_gb      = (known after apply)
  + name              = (known after apply)
  + storage_account_type = "Standard_LRS"
  + write_accelerator_enabled = false
}
```

```

+ source_image_reference {
  + offer      = "UbuntuServer"
  + publisher  = "Canonical"
  + sku        = "18.04-LTS"
  + version    = "latest"
}

+ termination_notification (known after apply)
}

# azure_rm_network_interface.nic_demo will be created
+ resource "azure_rm_network_interface" "nic_demo" {
  + accelerated_networking_enabled = (known after apply)
  + applied_dns_servers            = (known after apply)
  + dns_servers                   = (known after apply)
  + enable_accelerated_networking = (known after apply)
  + enable_ip_forwarding          = (known after apply)
  + id                            = (known after apply)
  + internal_domain_name_suffix   = (known after apply)
  + ip_forwarding_enabled         = (known after apply)
  + location                     = "spaincentral"
  + mac_address                   = (known after apply)
  + name                          = "nic-tf-lab-joao"
  + private_ip_address            = (known after apply)
  + private_ip_addresses          = (known after apply)
  + resource_group_name           = "rg-tf-lab-joao"
  + virtual_machine_id            = (known after apply)

  + ip_configuration {
    + gateway_load_balancer_frontend_ip_configuration_id = (known after apply)
    + name                                               = "ipconfig1"
    + primary                                           = (known after apply)
    + private_ip_address                               = (known after apply)
    + private_ip_address_allocation                     = "Dynamic"
    + private_ip_address_version                       = "IPv4"
    + subnet_id                                         = (known after apply)
  }
}

# azure_rm_resource_group.rg_demo will be created
+ resource "azure_rm_resource_group" "rg_demo" {
  + id      = (known after apply)
  + location = "spaincentral"
  + name    = "rg-tf-lab-joao"
}

```

```

# azure_rm_subnet.subnet_demo will be created
+ resource "azure_rm_subnet" "subnet_demo" {
  + address_prefixes = [
    + "10.0.1.0/24",
  ]
  + default_outbound_access_enabled = true
  + enforce_private_link_endpoint_network_policies = (known after apply)
  + enforce_private_link_service_network_policies = (known after apply)
  + id                                            = (known after apply)
  + name                                         = "subnet-tf-lab-joao"
  + private_endpoint_network_policies           = (known after apply)
  + private_endpoint_network_policies_enabled   = (known after apply)
  + private_link_service_network_policies_enabled = (known after apply)
  + resource_group_name                         = "rg-tf-lab-joao"
  + virtual_network_name                       = "vnet-tf-lab-joao"
}

# azure_rm_virtual_network.vnet_demo will be created
+ resource "azure_rm_virtual_network" "vnet_demo" {
  + address_space = [
    + "10.0.0.0/16",
  ]
  + dns_servers    = (known after apply)
  + guid           = (known after apply)
  + id            = (known after apply)
  + location       = "spaincentral"
  + name          = "vnet-tf-lab-joao"
  + resource_group_name = "rg-tf-lab-joao"
  + subnet         = (known after apply)
}

```

Plan: 5 to add, 0 to change, 0 to destroy.

We have the successful terraform apply:

```
Plan: 5 to add, 0 to change, 0 to destroy.
azure_rm_resource_group.rg_demo: Creating...
azure_rm_resource_group.rg_demo: Still creating... [00m10s elapsed]
azure_rm_resource_group.rg_demo: Creation complete after 14s [id=/subscriptions/daf9c53c-7096-4293-9bb1-f7ad8263db1a/resourceGroups/rg-tf-lab-joao]
azure_rm_virtual_network.vnet_demo: Creating...
azure_rm_virtual_network.vnet_demo: Creation complete after 7s [id=/subscriptions/daf9c53c-7096-4293-9bb1-f7ad8263db1a/resourceGroups/rg-tf-lab-joao/providers/Microsoft.Network/virtualNetworks/vnet-tf-lab-joao]
azure_rm_subnet.subnet_demo: Creating...
azure_rm_subnet.subnet_demo: Creation complete after 8s [id=/subscriptions/daf9c53c-7096-4293-9bb1-f7ad8263db1a/resourceGroups/rg-tf-lab-joao/providers/Microsoft.Network/virtualNetworks/vnet-tf-lab-joao/subnets/subnet-tf-lab-joao]
azure_rm_network_interface.nic_demo: Creating...
azure_rm_network_interface.nic_demo: Still creating... [00m10s elapsed]
azure_rm_network_interface.nic_demo: Creation complete after 13s [id=/subscriptions/daf9c53c-7096-4293-9bb1-f7ad8263db1a/resourceGroups/rg-tf-lab-joao/providers/Microsoft.Network/networkInterfaces/nic-tf-lab-joao]
azure_rm_linux_virtual_machine.vm_demo: Creating...
azure_rm_linux_virtual_machine.vm_demo: Still creating... [00m10s elapsed]
azure_rm_linux_virtual_machine.vm_demo: Still creating... [00m20s elapsed]
azure_rm_linux_virtual_machine.vm_demo: Still creating... [00m30s elapsed]
azure_rm_linux_virtual_machine.vm_demo: Still creating... [00m40s elapsed]
azure_rm_linux_virtual_machine.vm_demo: Still creating... [00m50s elapsed]
azure_rm_linux_virtual_machine.vm_demo: Creation complete after 51s [id=/subscriptions/daf9c53c-7096-4293-9bb1-f7ad8263db1a/resourceGroups/rg-tf-lab-joao/providers/Microsoft.Compute/virtualMachines/vm-tf-lab-joao]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.
```

VM running:

Connect

Connect via Bastion

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Open in mobile

Feedback

CLI / PS

Resource group (move)

Status

Location

Subscription (move)

Subscription ID

rg-tf-lab-joao

: Running

: Spain Central

: Diogo's Subscription

: daf9c53c-7096-4293-9bb1-f7ad8263db1a

Operating system

Size

Primary NIC public IP

Virtual network/subnet

DNS name

Health state

Time created

: Linux (ubuntu 18.04)

: Standard F2 (2 vcpus, 4 GiB memory)

: ☐

: vnet-tf-lab-joao/subnet-tf-lab-joao

: ☐

: -

: 18/05/2026, 7:17 am UTC

Tags (edit)

: Add tags

Properties

Monitoring

Capabilities (7)

Recommendations

Tutorials

Virtual machine

Computer name

Operating system

VM generation

VM architecture

Agent status

Agent version

Hibernation

Host group

Host

Proximity placement group

vm-tf-lab-joao

Linux (ubuntu 18.04)

V1

x64

Ready

2.15.0.1

Disabled

-

-

-

Networking

Public IP address

Public IP address (IPv6)

Private IP address

Private IP address (IPv6)

Virtual network/subnet

DNS name

-

-

10.0.1.4

-

vnet-tf-lab-joao/subnet-tf-lab-joao

-

Size

Size

vCPUs

Standard F2

2

4 GiB

mi8dgramm_SFm_k_Ci0KCOiwiJvQORhCYARiFAMits3Jty90kDQKleKURitMw8HINenNfiGwYqblK37zt53AvdPnWWB418AK0aAiQrFAIw_wcB_k_#